

SUMMARY

- Highly accomplished and results-driven Python Developer with 5 years of experience in designing, developing, and deploying robust and scalable backend solutions within cloud-based environments.
- Proven expertise in building and optimizing ETL pipelines, automating data workflows, and managing cloud infrastructure using AWS services. Proficient in leveraging Python frameworks like Django and FastAPI to develop high-performance RESTful APIs and microservices architectures.
- Progressively advanced from data automation and analytics to developing Generative AI-driven solutions, integrating LangChain, OpenAI API, and RAG pipelines to enhance backend intelligence and enable context-aware system automation.
- Tableau Experience including analysis, modeling, design, and development of Tableau, reports and dashboards for analytics and reporting applications.
- Experienced in developing high-throughput ETL and data processing pipelines handling millions of records per cycle, optimizing PostgreSQL and Python multiprocessing for performance, scalability, and reliability in production environments.
- Skilled in designing and deploying modular microservices and automated workflows using Argo Workflows, Docker, and REST APIs, ensuring consistent deployments and seamless integration across large-scale distributed systems.
- Strong DevOps background with experience in implementing CI/CD pipelines, automating infrastructure provisioning, and monitoring system performance to ensure reliability and efficiency.
- Passionate about leveraging cutting-edge technologies to enhance data processing, improve application performance, and drive innovation within dynamic and fast-paced environments.
- Deep knowledge of AWS services like EC2, S3, RDS, Lambda, and Docker, ensuring scalable, secure, and highly available cloud infrastructures that meet the demands of enterprise-level applications and workflows.
- Strong focus on performance optimization and automation, implementing solutions that enhance system throughput, reduce manual intervention, and minimize overall operational complexity for seamless scalability.
- Adept in implementing and managing data security measures, ensuring that all systems adhere to industry compliance standards across cloud environments, safeguarding sensitive data, and protecting against vulnerabilities.
- Extensive hands-on experience in managing and optimizing data pipelines using AWS Glue, Apache Airflow, and other ETL tools, ensuring efficient processing, transformation, and movement of large datasets in cloud environments.
- Experienced in collaborating within cross-functional teams, mentoring junior developers, and leading Agile development practices to ensure high-quality deliverables and fostering a culture of continuous learning.
- Proficient in test-driven development (TDD) and automated testing frameworks, ensuring the delivery of robust, reliable, and maintainable code that adheres to strict quality standards across various stages of development.
- Proven track record in troubleshooting and resolving complex technical issues, consistently delivering high-performance solutions to meet the evolving needs of mission-critical operations in dynamic environments.

SKILLS & INTERESTS

Programming Languages: Python, JavaScript, SQL

Frameworks: Django, Flask

AI & Generative Systems: LangChain, OpenAI API (GPT-4), Retrieval-Augmented Generation (RAG), FAISS, Pinecone, Prompt Engineering, Embedding Models, LLM Integration, Hugging Face Transformers

API Development: REST, SOAP, FastAPI

Databases: MySQL, PostgreSQL, NoSQL (Cassandra, MongoDB)

Cloud Platforms: AWS (EC2, S3, RDS, Lambda), GCP (Compute Engine, Cloud SQL, BigQuery)

CI/CD Tools: Jenkins, GitLab CI, Docker

Testing Tools: PyUnit, Selenium

Data Tools: Pandas, Elasticsearch, Kibana, Grafana, Tableau

Web Technologies: HTML, CSS, JavaScript, ReactJS, RAPID, TypeScript/XML, PromQL, GraphQL

Version Control: Git ETL

Tools: AWS Glue, Apache Airflow, Snowflake, Debugging, Argo Workflows, ConfigMap, Terraform, CloudFormation, Kubernetes, SAP APIs, Multiprocessing

WORK EXPERIENCE

Bloomberg

Full Stack Developer

Nov 2024 – Present

- Developed and optimized ETL pipelines using Python, Pandas, and AWS Glue to automate data ingestion, transformation, and loading processes, improving data workflow efficiency and scalability in cloud-based environments.
- Re-engineered FXGO SAP Metering workflows into modular Python-based microservices, improving scalability, maintainability, and performance across trade validation, aggregation, and billing systems.

- Designed and implemented high-throughput ETL pipelines using Python multiprocessing and PostgreSQL batch operations, processing over 2M+ trade records per cycle with optimized transaction handling and buffered logging.
- Developed and maintained data ingestion and transformation workflows, ensuring data accuracy, consistency, and integrity across multiple environments and large-scale datasets.
- Integrated SAP REST APIs using Bloomberg BMS Service Discovery for dynamic endpoint management, secure authentication, and seamless synchronization of pricing and tier-based data across environments.
- Implemented data auditing and validation mechanisms within pipelines to ensure transparency and traceability for compliance and internal reporting processes.
- Built and maintained RAPID-based dashboards in TypeScript/XML to visualize live billing, workflow status, and incentive data directly from PostgreSQL, enabling real-time monitoring and operational decision-making.
- Automated full lifecycle deployments using Argo Workflows and ConfigMap scripts, streamlining environment-specific configurations and ensuring consistency across development, staging, and production systems.
- Enhanced PostgreSQL query performance through indexing, optimized joins, and connection pooling, reducing query execution time and improving overall data pipeline throughput.
- Collaborated with cross-functional teams to implement structured data schemas and maintain data lineage across ETL processes, improving data accessibility and reliability for analytics and reporting.
- Coordinated with DevOps and data engineering teams to monitor pipeline health, manage version control, and ensure reliable, secure, and timely data delivery across all integrated systems.
- Designed and implemented an AI-powered analytics assistant using LangChain + GPT-4 + internal APIs, enabling automated query generation and reporting on trade and billing data.
- Developed retrieval pipelines combining PostgreSQL and FAISS embeddings to support context-aware insights within ETL workflows, reducing manual analysis effort by 35%.
- Led evaluation of prompt engineering strategies and RAG architectures for integrating GenAI components into Bloomberg's existing microservices ecosystem, ensuring security, latency, and compliance standards.
- Collaborated with infrastructure teams to containerize and deploy AI microservices via Argo Workflows + Docker, ensuring scalability and reproducibility in production.

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Sr. Software Engineer

Aug 2023 – Oct 2024

- Developed and optimized ETL pipelines using Python, Pandas, and AWS Glue to automate data ingestion, transformation, and loading processes, improving data workflow efficiency and scalability in cloud-based environments.
- Designed, developed, and deployed RESTful APIs for seamless system-to-system data integration, enhancing application interoperability and reducing latency in distributed architecture environments.
- Managed AWS infrastructure, including EC2, S3, and RDS, optimizing resource allocation and implementing best practices for security and access management, ensuring efficient and secure cloud operations.
- Designed and developed web services, micro services deployed/running on cloud technologies like Pivotal Cloud Foundry (PCF) and AWS (Amazon Web Services). Created and published REST web services using AWS API gateway. Deployed application as a service on AWS EC2(Elastic Compute Cloud) using kubernetes and docker. Used AWS CloudWatch for monitoring health and performance of application services and AWS CloudTrail for verifying logs and debugging any application errors.
- Automated CI/CD pipelines with Jenkins, Docker, and GitLab, streamlining build, testing, and deployment workflows, ensuring consistent code quality, faster release cycles, and improved system reliability.
- Collaborated with cross-functional teams to lead DevOps initiatives, implementing infrastructure automation and monitoring solutions to improve system uptime, reduce incident response times, and enhance overall system performance.
- Experience in tableau created Rapid Analytics and Reporting.Created SQL scripts for testing and validating data on various reports and dashboards and handled performance issues effectively in Tableau.
- Enhanced application performance by optimizing SQL queries, indexing strategies, and implementing advanced caching mechanisms, significantly improving data retrieval times and reducing overall system load.
- Optimized Tableau report performance by reducing load time through efficient use of indexed data sources, optimizing calculated fields, minimizing table joins, and applying context filters. Implemented extract-based data connections and optimized workbook structures to enhance query execution efficiency.
- Developed custom Grafana panels and fine-tuned PromQL queries to visualize high-resolution time-series data across distributed environments. Leveraged multi-dimensional data filtering and dynamic variables to monitor system logs, network latency, and resource usage, providing detailed insights into infrastructure performance and bottlenecks.
- Conducted technical trainings for team members. Assisted stakeholders and functional teams with technical/functional queries. Defect fixing and defect management using tools like JIRA, Quality Control etc.
- Worked closely with data science teams to deploy machine learning models into production environments, integrating real-time data streams and enabling actionable insights for business decision-making and predictive analytics.
- Partnered with data science teams to deploy and monitor LLM-based summarization and tagging models for large image-metadata datasets, improving search relevance and content classification accuracy.
- Built an internal developer assistant using LangChain + OpenAI API, integrated with internal APIs and documentation to reduce query resolution time for engineers.
- Developed embedding pipelines using FAISS and Pinecone for semantic search over media assets, enhancing retrieval precision and contextual indexing across millions of records

- Built and automated resilient ETL pipelines using Python and AWS Lambda, optimizing data extraction, transformation, and loading across cloud systems, ensuring scalability and seamless integration with distributed data workflows.
- Engineered and managed MySQL and PostgreSQL databases, enabling efficient data storage and processing for large-scale applications, while ensuring data integrity, optimized query performance, and high availability.
- Developed RESTful and FastAPI-based backend services to automate workflows and enable efficient data exchange between cloud services and internal systems, improving cross-platform interoperability and reducing latency.
- Deployed and maintained scalable cloud-based storage solutions using AWS EC2, S3, and RDS, optimizing performance, ensuring high availability, and delivering cost-efficient infrastructure for large-scale, data-driven applications.
- Automated cloud infrastructure provisioning using CloudFormation and Terraform, significantly reducing manual intervention, improving deployment speed, and ensuring consistent, reliable resource allocation across AWS environments.
- Designed and implemented microservices architecture with FastAPI, enhancing the scalability, fault tolerance, and maintainability of backend services, supporting high-volume, distributed data processing workflows.
- Designed and implemented GraphQL APIs, enabling clients to fetch data in a flexible and efficient manner, reducing over-fetching and under-fetching issues common in REST APIs.
- Optimized query performance in GraphQL by leveraging efficient data fetching strategies, such as batching and caching, to improve API response times and reduce server load.
- Streamlined DevOps processes by optimizing CI/CD pipelines using Jenkins and Docker, automating the build, testing, and deployment cycles, leading to faster, error-free releases and improved system stability.
- Configured and monitored AWS CloudWatch and Elasticsearch for real-time logging, application performance monitoring, and proactive issue detection, ensuring system reliability and early identification of potential problems.
- Implemented high-performance backend services using FastAPI, optimizing request handling and reducing response times, ensuring efficient API interactions and enhancing the overall application throughput in cloud environments.
- Developed a Retrieval-Augmented Generation (RAG) prototype combining LangChain + FAISS + FastAPI, enabling contextual search and document-aware question answering.
- Integrated OpenAI API endpoints for internal experimentation, exploring prompt design and response optimization for automating data documentation and reporting workflows.
- Designed reusable API wrappers to standardize LLM inference calls and error handling for downstream applications.

- Supported the development and deployment of ETL pipelines utilizing AWS services like Lambda, Glue, and S3, enhancing data integration capabilities and optimizing workflows for efficient data processing and engineering operations.
- Automated routine data transformation tasks with Python, significantly improving workflow efficiency, reducing manual intervention, and ensuring consistency and reliability in data processing pipelines across cloud environments.
- Deployed and managed Docker containers to maintain consistent environments across development, staging, and production systems, minimizing deployment discrepancies and enhancing the reliability of code execution.
- Managed Git workflows, overseeing repository maintenance and code versioning to improve collaboration, streamline development processes, and ensure seamless integration and deployment cycles.
- Worked closely with senior developers to enhance DevOps workflows by automating infrastructure deployment, improving system reliability, minimizing downtime, and optimizing resource management through continuous improvements.
- Implemented and managed CI/CD pipelines using GitLab CI and Jenkins, automating testing, integration, and deployment processes to facilitate smoother development operations and faster release cycles.
- Built and deployed high-performance backend services using FastAPI, improving API response times and overall system throughput for rapid, scalable backend development in cloud environments.
- Collaborated with frontend teams to enable precise data fetching using GraphQL fragments, enhancing application performance and improving developer efficiency.
- Collaborated on optimizing cloud infrastructure by leveraging AWS ECS and Lambda, deploying microservices within containers and serverless environments to ensure scalability and resource efficiency.
- Monitored application performance using AWS CloudWatch and Elasticsearch, enabling proactive issue detection, ensuring high system uptime, and facilitating rapid resolution of performance bottlenecks.
- Collaborated with data engineering teams to prepare and clean datasets for machine learning model training and evaluation, leveraging AWS Glue and Lambda for automated preprocessing.
- Gained hands-on exposure to Python ML libraries (scikit-learn, Pandas) for exploratory data analysis and feature validation in internal proof-of-concept automation tools.

- Assisted in the design and implementation of Django-based web applications, collaborating with senior developers to ensure architecture and functionality align with business and project requirements for scalable, maintainable systems.

- Contributed to the development of RESTful APIs for efficient data retrieval and manipulation, integrating them into legacy systems to improve data flow and enable smoother, real-time data processing and cross-application communication.
- Conducted data analysis and wrote Python scripts for cleaning and preprocessing large datasets, improving data quality and accuracy for reporting, analytics, and decision-making, ensuring optimized data workflows.
- Collaborated in the debugging process, identifying, diagnosing, and resolving application-level code issues, ensuring smooth performance and minimizing downtime while adhering to best practices in code optimization and error handling.
- Coordinate hardware/software configurations and upgrades that impact daily payment processing. Manage both large and small-scale projects using MS Project as management tool to track overall project status identify impediments as well as bottle necks.
- Experimented with simple Python-based text analytics and keyword extraction scripts to automate internal reporting tasks, forming the foundation for later interest in data-driven and AI-assisted systems.

CERTIFICATIONS

- Amazon Web Services Cloud Practitioner
- Certified Training in C and C++ Programming Languages

EDUCATION

- Master of Science in Computer Science [University of Massachusetts]